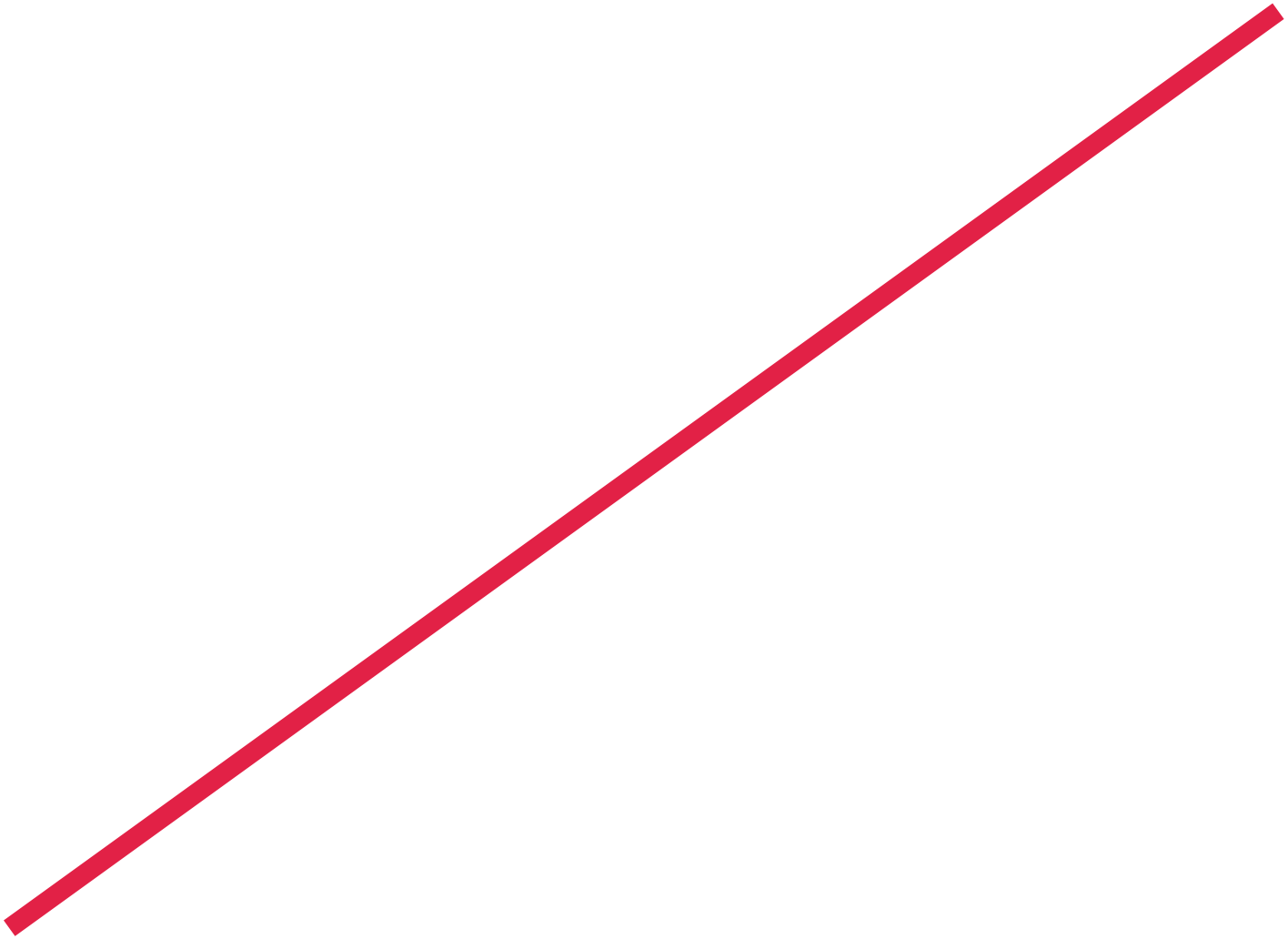


Q1: How many solutions?

$$ax + b = 0$$









a

$\neq$

O

$$a = 0, \quad b \neq 0$$

a = 0, *b* = 0

Number of solutions



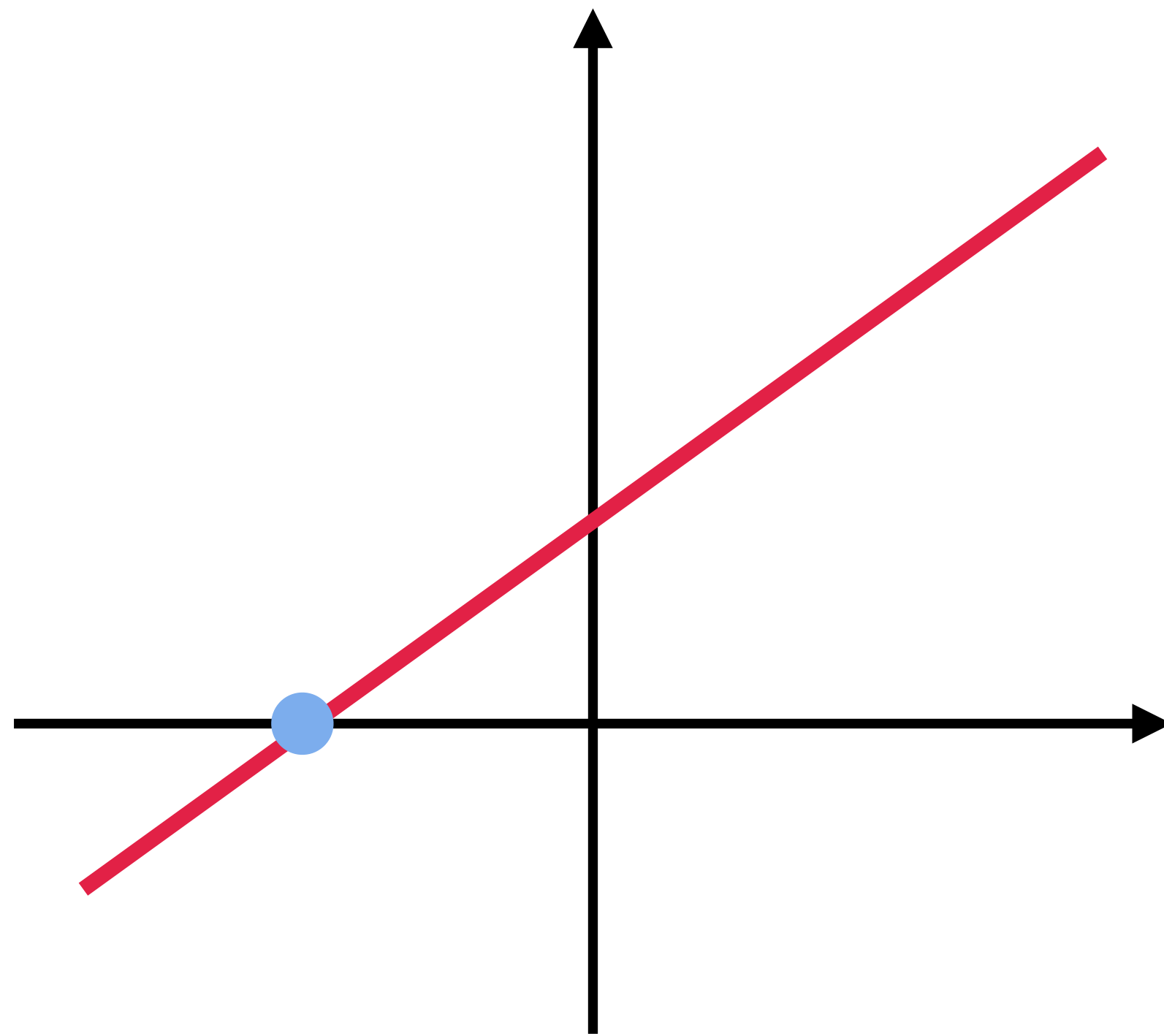




Bad cases!

Q1: How many solutions?

$$ax + b = 0$$



Number of solutions

$$a \neq 0$$

1

$$a = 0, \quad b \neq 0$$

0

$$a = 0, \quad b = 0$$

∞

Bad cases!

Q1: How many solutions?

Problem: Parallel lines don't intersect!
